

Detection of Different Types of Ear Diseases in Infants using Deep Learning for Early Treatment

Sindu Divakaran^{1*}, T Sudhakar¹, Lumen Christy V¹, Lekha Parthiban¹, Fahira Parveen M S¹,
Berti Maria Preeta¹, S Krishnakumar¹

¹Department of Biomedical Engineering, School of Bio and Chemical Engineering,
Sathyabama Institute of Science and Technology, Chennai, Tamilnadu-600119

[*sindudiva@gmail.com](mailto:sindudiva@gmail.com)

Abstract

The loss or inability of auditory sense is the most common sensory organ deficiency. Out of every 1000 new-borns, 5-6 are deaf or hard of hearing. They will not be detected until they are two or three years old, after which permanent trauma would have happened. One of the most prominent reasons for deafness in infants is due to ear diseases. Early detection of ear diseases is the top priority to prevent deafness in infants. This research work has developed and applied modified deep learning architecture such as AlexNet, mini GoogLeNet, and LeNet and compared the training by the three architectures. Of the three, AlexNet architecture detected and identified the presence of different ear ailments such as acute otitis media, glue ear, safe CSOM, otomycosis, other infections and also checked whether the ear is normal, with high accuracy.

Keywords: Heard of hearing, ear diseases, AlexNet, mini GoogLeNet, LeNet

1. INTRODUCTION

Human perception of the environment is mediated by sensory perceptions. Hearing is the sense that allows people to form relationships, engage in everyday activities, be alerted to risk, and encounter life events. (Abu Naser et.al.,2016) Globally, the fourth leading cause of disability is loss of hearing. (Cunningham et.al.,2017) Around 466 million people, or 5% of the global population, have hearing damage that is considered disabling; about 34 million of these are children. (Neumann et.al.,2019) The vast majority of people live in economically developing countries around the world. (Hapunda et.al.,2020) Hearing damage in children can be caused by a variety of ear disorders. An ear infection can affect anyone at any age, but it is more common in children than in adults since a child's immune system is less strong. 5 percent of children have congenital auricular deformities. (Hallac et.al.,2019) This include impacted cerumen (excessive ear wax) and glue ear (non-suppurative otitis media), which is caused by fluid deposition within the ear. According to the World Health

Organization (WHO), preventative interventions could prevent about 60% of childhood hearing loss. (WHO 2016)

Ear infections in children may also result delayed speech and language development as well as academic and educational development. (Rosenfeld et.al.,2014) In order to prevent the deafness caused by ear diseases, it is necessary to diagnose these diseases at an early stage. There are many ways to scan an ear but there is no proper way to detect the type of disease in the ear. Errors might occur when diagnostic tools are used by technicians who are not specialised in otology. (Dongchul et.al.,2019) The work determines the occurrence of ear disease in an infant, to reduce the time required to expose the existence of ear diseases in infants and to determine the cheap way to predict the presence of ear diseases. In order to meet these goals, a bibliographic review was conducted to analyse the different types of ear disease detection techniques and methods that are available.

Deep learning is a form of machine software that simulates the brain's network of neurons. It is a branch of machine learning which is referred to as deep learning because it employs deep neural networks. Convolutional Neural Network (CNN) is a multi-layered neural network with a special architecture that determines the performance by extracting increasingly complex features from the data at each layer. By training the images with the CNN architectures, the ear diseases could be predicted at an early stage with high accuracy. Advantages includes Early detection of diseases, good accuracy, Human error rate in diagnosis can be eliminated, cost effective and time saving. Practical reasons include, Robot companion for auditory rehabilitation, Ear condition diagnosis and differentiate ear diagnosis.

Dataset of five different types of ear diseases was collected namely acute otitis media, safe chronic suppurative otitis media, glue ear, otomycosis, other types of ear infections along with normal ear images. Otomycosis is a fungal ear infection that

affects the outer ear canal and is caused by a superficial mycotic infection. Based on whether or not the illness is present, it may be classified as active or inactive. Adhesive otitis, also known as glue ear, is a condition in which the middle area of the ear is accumulated with fluid. Acute otitis media is a condition where the middle ear, which is located behind the eardrum, gets inflamed and sick. It is the second most common paediatric emergency after upper respiratory infections. (Lipo et.al.,2017)

This dataset was pre-processed and augmented. AlexNet, mini GoogLeNet and LeNet are employed to train the datasets. The output of the training done by these three architectures was compared to find out which architecture trained the images the best.

II. MATERIALS AND METHODS

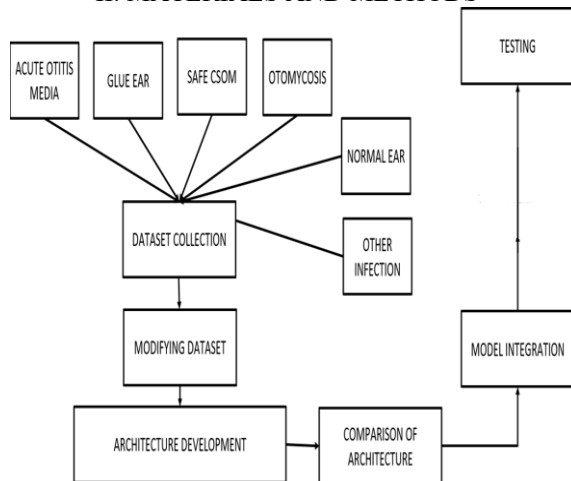


Figure 1 Work pattern of the system

Collection of Ear Disease Datasets

This step included collecting otoscopic images of the five forms of ear diseases as well as images of the normal ear from National Library of Medicine and few high clarity google images. (Viscaino et.al.,2020) The image dataset of the six forms of ear diseases was downloaded from multiple internet research sites. The consistency improves as the dataset size grows. The dataset was divided into training and testing data subsets after the images were obtained.

Image Pre-processing

Undesired distortions are removed to improve picture data and increase key visual qualities that are important for further processing and analysis. Usually, the image dataset will not be clean in most cases. The data cannot be used as training dataset for the model. These unclean datasets are resized and made use for further processing.

Image Augmentation

Image data augmentation is a method for artificially increasing the size of a training dataset by modifying images in the dataset. Image

augmentation artificially produces training images using a variety of imaging techniques or a mixture of techniques, such as spontaneous rotation, shifts, shear, and flips, among others. Deep learning neural network models that are trained on more data become more skilled, and augmentation techniques may generate variations of the images that enhance the fit models' ability to generalise what they've learned to new images

Convolutional Neural Networks

Machine learning algorithms for medical image processing, stressing clinical aspects of the field concentrates on convolutional neural networks. The convolutional framework, which is made up of convolution and pooling layers, and a series of completely linked layers on top that performs the classification operation, make up the CNN architecture. Each layer is made up of a group of neurons, each of which is completely connected to all neurons in the layer before it. Finally, the output layer, which represents the predictions, is the fully connected layer. In this work, we employ three CNNs, AlexNet, MiniGoogLeNet and LeNet to train the dataset and in predicting the results.

AlexNet

The network can take the input image having height, width as multiples of 32 and 3 as channel width. It consists of multiple layers of convolution, pooling, activation functions and dropout functions.

Mini GoogLeNet

Inception architecture is the platform on which GoogLeNet is built. It employs Inception modules, which enable the network to choose from a variety of convolutional filter sizes in each block. Dimensionality reduction was used to make the architecture computationally efficient.

LeNet

CNN models differ from LeNet models in several ways, but they are all built on LeNet. LeNet contains the basic components of a convolutional neural network laying the groundwork for potential convolutional neural network growth. There are seven levels of LeNet.

Comparison of Architecture

After training the dataset using three different architecture comparison has to be made among them to find out the architecture with the best training accuracy. This comparison was done based on certain parameters like accuracy, precision, recall, f1-score, weighted average and macro-average.

III. RESULTS AND DISCUSSION

Detection procedure:

Collection of datasets

In the first step otoscopic images of different ear diseases including acute otitis media, safe CSOM, glue ear, otomycosis, other types of ear infections and images of normal ear. Figure 2 is a sample of the dataset collected. 300 images have been

collected in total, 50 for each type of disease from different websites such as Kaggle and dataworld.

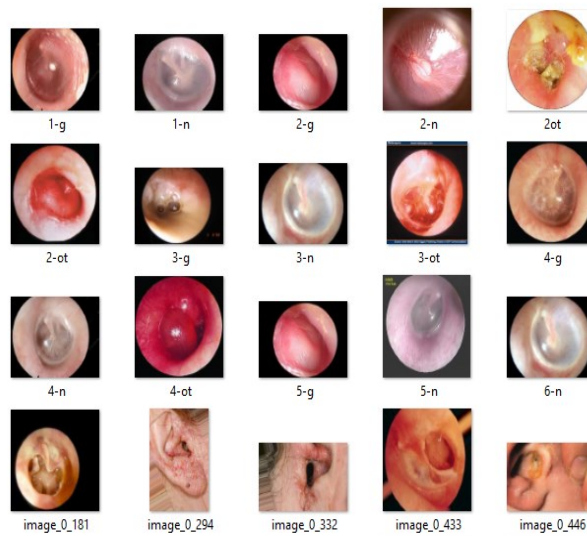


Figure 2 Collected Dataset

After collection of datasets, the datasets are separated into different categories of diseases.

Pre-processing output:

The dataset collected undergo pre-processing to optimize image data by suppressing unwanted distortions or enhancing certain image features that are essential for subsequent processing. It is important that images are brought into uniform dimensions.

Augmentation output

The next step is data augmentation, in this phase the images collected are set to standard resolution format. Figure 3 show the image before augmentation. Image augmentation artificially produces training images using a variety of imaging techniques or a mixture of techniques, such as spontaneous rotation, shifts, shear, and flips, among others. For this purpose, class ImageDataGenerator is used. Using real-time data augmentation, it creates minibatches of image data. At each epoch, the ImageDataGenerator class ensures that the model receives new variations of the images. The image dataset will not be clean in most cases and hence it cannot be used as training data for this reason the data is brought to standardized form by resizing it. This can be done using augmentsAware Preprocessor (for resizing the image) and Image to Array Preprocessor (to align standard height width and depth). Figure 4 shows the images after augmentation. It is clearly shown that giving one input image has generated 10 output images that are uniform in dimensions.

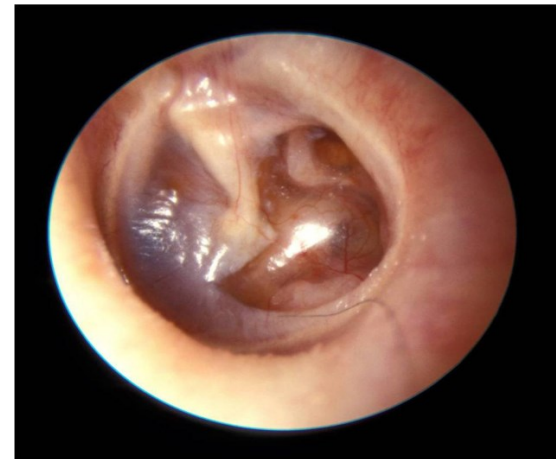


Figure 3 Image before augmentation



Figure 4 Image after augmentation

Training Outputs

From the images obtained, 75% have been used for the training and 25% have been used for testing.

AlexNet Output

Figure 5 shows the graph before fine tuning the CNN. From the graph, it can be clearly seen that the network doesn't predict all the six categories of diseases with 100% accuracy. For certain diseases like glue-ear and otomycosis the values obtained are very less.

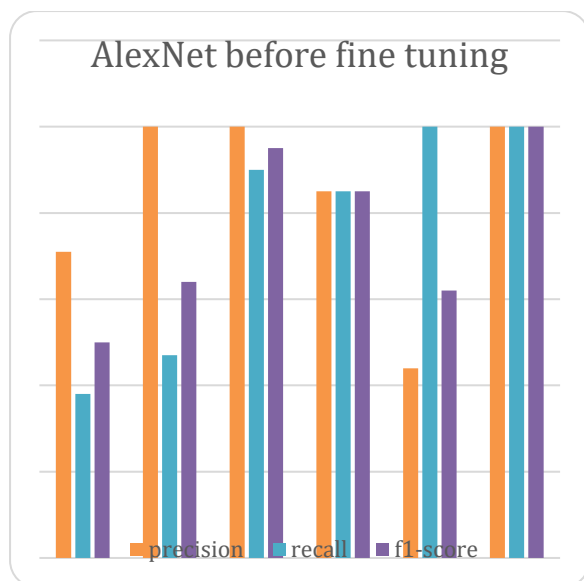


Figure 5 AlexNet training graph

Fine tuning was carried out in order to improve the architecture's correctness, where the datasets were trained for an increased number of epochs. Figure 6 shows the output obtained after fine tuning. The accuracy has raised from 0.75 to 1.00 after fine tuning. That is the architecture has been trained to detect the diseases with 100% accuracy. From the values obtained, it is inferred that the network has trained the architecture very effectively and it is very efficient in predicting the six categories of images.

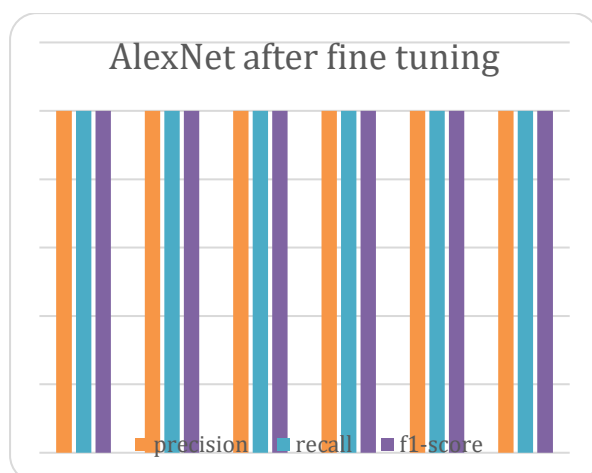


Figure 6 AlexNet training graph after fine tuning

Figure 7 shows the graph that is obtained after the training. It is the training loss and accuracy graph.

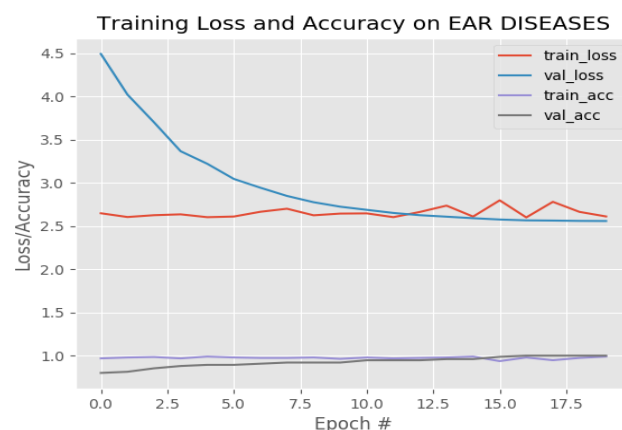


Figure 7 AlexNet loss/accuracy Graph

The percentage of precise predictions for the test results is known as accuracy. It's easy to figure out by calculating the number of valid predictions by the total number of predictions.

$$accuracy = \frac{\text{correct predictions}}{\text{all predictions}}$$

The training loss has started around 2.6 and after fluctuating a little around 13-17 epochs it has returned to its original value. The validation loss came down to 2.5 while starting at 4.5. The training accuracy has been stable at 1.0 while the validation accuracy has started at 0.9 at gradually stabilised to 1.0.

Mini GoogLeNet Output

The accuracy value is 0.25. The precision, recall, and f1-score of almost all diseases is zero, with the values of normal ear images and otomycosis being slightly higher. Figure 8 is the graphical representation of the values obtained from the training of the dataset by Mini GoogLeNet.

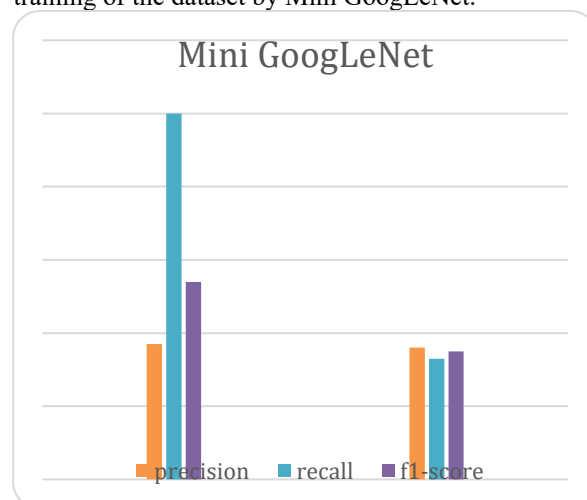


Figure 8 Mini GoogLeNet training graph

Figure 9 shows the training loss accuracy graph of GoogLeNet architecture.

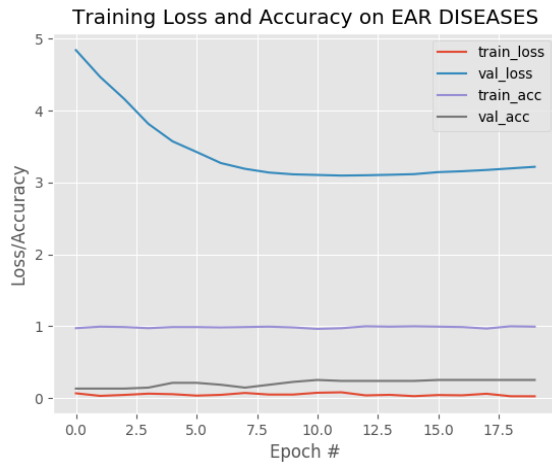


Figure 9 Mini GoogLeNet loss/accuracy graph

The training loss has been stable at 0. The validation loss reduces to 3 from 5. The training accuracy has been stable at 1.0 but the validation accuracy has been around 0. The training by Mini GoogLeNet is very poor and not as efficient as AlexNet.

LeNet Output

The precision, recall and f-score of all the six categories has been found to be in between 0.5 and 1.0. Among all the diseases, the value of other infection is 1 and the lowest value has been obtained in case of otitis media, which means that the network identifies other infections very effectively but it identifies otitis media with less accuracy. Figure 10 is the graphical representation of the values obtained from the training of the dataset by LeNet. The values obtained clearly indicated that the training by LeNet architecture is much better compared to the training by Mini GoogLeNet.

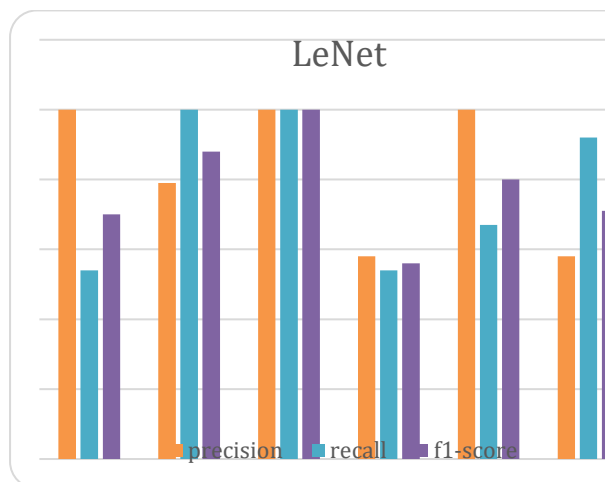


Figure 10 LeNet training graph

Figure 11 shows the training loss accuracy graph of LeNet. The training loss has started at 0.775 and it has been fluctuating throughout.

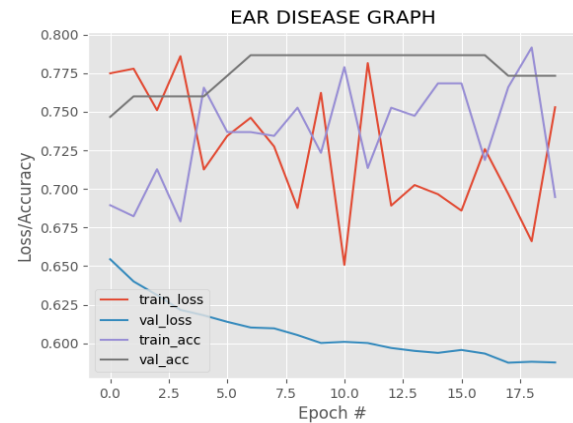


Figure 11 LeNet loss/accuracy graph

The validation loss reduces to 0.600 from 0.650. The training accuracy has started at 0.690 and has been fluctuating throughout. The validation accuracy has been started at 0.750 and at the end of 20 epochs, it is at 0.775. Here, the accuracy is higher than the Mini GoogLeNet but it is less compared to AlexNet. The accuracy value is 0.77.

Comparison of architecture

After training the dataset by the three architectures, using the training outputs generated, a comparison was made between the three architectures to find which architecture trained the dataset the best. Table 1 summarizes the parameters that has been used to compare the architectures and the values that has been obtained

From the above values the Detection rate using these models has been significantly improved by

Table 1: Comparison of architectures

Parameters	ALEXNET	MINI GOOGLNET	LENET
Accuracy	1.00	0.25	0.77
Macro Average	1.00	0.15	0.78
Weighted Average	1.00	0.16	0.77
F1 score	1.00	0.16	0.77
Precision	1.00	0.13	0.77
Recall	1.00	0.25	0.78

increasing the accuracy rate. It was concluded that AlexNet architecture trains the images the best and it is used in the back-end. Comparative lower values with respect to F1 score, precision, macro and weighted average, accuracy and recall leads to an interpretation that LeNet and mini GoogLeNet are not able to detect features precisely.

IV. CONCLUSION

This research study has successfully developed and applied three different modified deep learning architectures namely, AlexNet, mini GoogLeNet and LeNet to determine and classify the presence of different types of ear diseases. The data is obtained, modified and trained using the deep learning networks. Among the three architectures compared, AlexNet emerged with higher accuracy. This initiative has a bright future ahead of it, where manual prediction can be easily transformed into computerized prediction at a low cost.

REFERENCES

- [1] Samy S. Abu Naser, Hasan A. Abu Hasanein , “Ear Diseases Diagnosis Expert System Using SL5 Object”, Research Gate, Vol 2, January 2016, 41-47.
- [2] [Joanne E Roberts](#), [Richard M Rosenfeld](#), [Susan A Zeisel](#), “Otitis media and speech and language: a meta-analysis of prospective studies”, *PubMed Central*, Vol. 113(3), March 2014, 238-48.
- [3] Katrin Neumann, Shelly Chadha, George Tavartkiladze, Xingkuan Bu, Karl L White, “Newborn and Infant Hearing Screening Facing Globally Growing Numbers of People Suffering from Disabling Hearing Loss:. *Intl Journal of Neonatal Screening*. January 2019, 5(1):7.
- [4] Hapunda R K, Aswani J, Kipinger M, and Munthali J, “Prevalence of Hearing Loss in Primary School Children in Lusaka, Zambia”, *Medical Journal of Zambia*, **Vol. 47 (2)**, March 2022, 91 – 97.
- [5] Rami R. Hallac, Jeon Lee, Mark Pressler , James R. Seawood and Alex A Kane, “Identifying Ear Abnormality from 2D Photographs Using Convolutional Neural Networks”, *Sci Rep*, Vol. **9**, 2019.

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Conflict of interest

The authors declare no conflict of interest regarding the work

- [6] Marwin Alejo, Cris Paulo Hate, “Unconstrained ear recognition through domain adaptive deep learning models of convolutional neural network”, *International Journal of Recent Technology and Engineering (IJRTE)*, Vol. 8, Issue-2, July 2019.
- [7] Sanchit Aggarwal, “Modern web-development using ReactJS”, *International Journal of Recent Research Aspects* ISSN: 2349-7688, Vol. 5, Issue 1, 2018, 133-137.
- [8] Mohammad Azam khan, Soonwook kwon et.al., “Automatic detection of tympanic membrane and middle ear infection from oto-endoscopic images via convolutional neural networks”, *Science Direct*, Vol. 126, June 2020, 384-394.
- [9] Dongchul Cha, Chongwon Pae et.al, “Automated diagnosis of ear disease using ensemble deep learning with a big otoendoscopy image database”, *eBioMedicine*, Vol. 45, July 2019, 606-619.
- [10] Xinyu Zeng, Zifan Jiang et.al, “Efficient and accurate identification of ear diseases using an ensemble deep learning model”, *Scientific Reports*, Vol. 11, May 2021.
- [11] Zheng Wong, Jian Sang et.al, “Structure-aware deep learning for chronic middle ear disease”, *ScienceDirect*, Vol. 194, May 2022.
- [12] Akshayaa Vaidyanathan, Marly F. J. et.al, “Deep learning for the fully automated segmentation of the inner ear on MRI” *Scientific reports*, Vol. 11, February 2021.